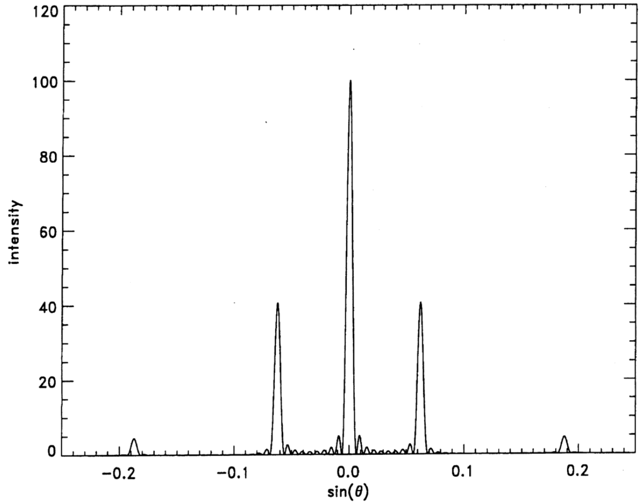
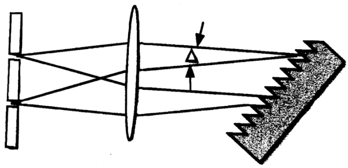
660 10. You have just taken a job as an optical designer for Aspheric Holistic Systems, whose motto is “expensive solutions for simple problems”. Your first assignment is to design a refractive lens that will focus monochromatic light from a small pinhole to a point at a distance of $2f$ with an image quality constrained only by diffraction considerations. The index of refraction of the lens material is n relative to air and the diameter of the lens must be $2R$. The lens will be placed symmetrically at the midpoint between pinhole and focus. Determine the equation for the two optical surfaces that will satisfy these requirements (It is NOT a sphere!). (Hint: take ruthless advantage of the symmetry of the problem).

661 6. A laser emitting radiation with wavelength, λ, in a broad parallel beam is directed to a target of equally spaced opaque lines of width 8λ separated by transparent intervals of width also 8λ. Sketch the intensity of the diffraction pattern and compute the angles corresponding to the first 2 prominent maxima on the positive side of the central intensity maximum.





662 7. A collimated laser beam has a Gaussian, TEM_{00}, transverse intensity profile,

$$I = I_0 e^{\frac{-2r^2}{w^2}}$$

with peak intensity of $I_0 = 10\,\mathrm{GW/cm^2}$ and width $w = 8$ mm. The laser is focused by a lens of diameter $d = 10$ cm and focal length 20 mm . The wavelength is $1.064\,\mu$m.
(a) What is the focused beam size, assuming an aberration free lens?
(b) What is the peak intensity at the focus?
(c) Compare the corresponding electric field with the electric field seen by an electron in a Hydrogen atom and comment on the result.

663 10. A grating monochromator is formed from a 1200 lines/mm grating and a single 250 mm focal length lens. The angular separation between the diffracted and incident beams, $\Delta = \theta_d - \theta_i$, is fixed by two closely spaced slits, the operational wavelength may be tuned by rotating the grating.

664 8. A variable optical attenuator is constructed by inserting a half-wave (retardation) plate between crossed linear polarizers. Describe the polarization state immediately after the waveplate when the fast axis makes an angle θ with respect to the second polarizer. What is the transmission through the second polarizer as a function of θ ?

665 10. A thin film of optical thickness $\lambda/4$ is coated on a glass substrate. In terms of the indices of refraction of the glass (n_2) and the surrounding media (n_0), determine the index of refraction n_1 of the film, such that there is no reflection of light of wavelength λ for illumination at normal incidence. *Hint: You may find the following sum useful*:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \text{ for } |x| < 1$$

666 8. Consider diffraction of plane monochromatic light through two vertical slits of width b separated by a distance $h = 3b$, where $b \gg \lambda$. Describe the diffraction pattern in the Fraunhofer limit. At what angles do you find the bright and dark fringes?

667 10. A collimated TEM_{00} Gaussian laser beam has a cylindrically symmetric intensity (irradiance) profile

$$I(r,t) = I_0(t)e^{-2r^2/\omega^2} (\omega = 2\sigma)$$

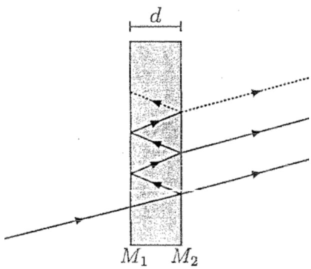
and a wavelength $\lambda = 1\mu$ m. If the peak intensity of the laser is 10GW/cm^2 for a width of $\omega = 1$ cm, what would the width and peak intensity be at the focus of a 30 cm lens (of sufficient diameter that there are negligible diffraction losses)? Compare the peak electric field seen at the focus, with the electric field seen by an electron orbiting in the ground state of a hydrogen atom.

668 6. A scuba diver begins a dive by swimming directly out from the diver's boat a distance of 50 m on the surface and then descends straight down toward the bottom. The index of refraction of water is $n = 1.3$. Assuming the water is perfectly transparent, and the diver is brightly lit by sunlight, at what depth will a friend in the boat first be able to see him under the water?

669 11. In a 2 -mirror (plane-parallel) Fabry-Perot etalon, the relative phase difference δ (in radians) between successive multiply-reflected waves is given by the optical path difference divided by the wavelength, times 2π. That is,

$$\delta = \frac{2\pi}{\lambda}(2nd\cos\theta)$$

where n is the refractive index of the medium between the mirrored interfaces M_1 and M_2, d is the mirror separation, and θ is the angle of incidence.



(a) Find the frequency separation $\nu_{m+1} - \nu_m$ of adjacent bright fringes in terms of n, d, θ, and c (the velocity of light).
(b) At normal incidence, for a fixed input wavelength and fixed n, what is the change in d that scans the device from one bright fringe to the next? That is, find $\Delta d = d_1 - d_2$ corresponding to one fringe.

670 6. An optical diffraction grating has many grooves on it separated by a constant distance h. The Frauenhofer diffraction pattern consists of bright fringes of different orders $m, m+1, m+2, \ldots$ that emerge at different angles with respect to the grating normal. Let the angle of the m^{th} -order bright fringe from the grating normal be θ.
(a) Derive the "grating" equation that relates the optical wavelength λ to m, h, and θ at intensity maxima.
(b) What is the angular separation $\Delta\theta$ of the intensity maxima for two different wavelengths λ_1 and λ_2 observed in the same order m ?

671 11. An "axicon" lens is constructed by cutting the tip of a smooth glass cone off, perpendicular to the cone axis along z. An optical element with two flat, circular faces is thereby formed (Fig. 1 below). The diameter of the cone base is b and the diameter of the face near the former tip of the cone is $a(a < b)$. Face a is opaque due to black paint applied as shown in Fig. 2. A parallel beam of light impinges on the b face from the left, travelling parallel to the z-axis, and rays farther off-axis than $a/2$ and less than $b/2$ refract at the second interface, where the index changes from $n_{\text{glass}} = 1.5$ to $n_{\text{air}} = 1.0$. If the cone angle is $\alpha = 120°$, and the dimensions are $a = 2$ cm, $b = 6$ cm and $h = 2/\sqrt{3} = 1.555$,
(a) find the angle β, and
(b) find the length f of the region where light is concentrated in a "line focus" after passing through the axicon.

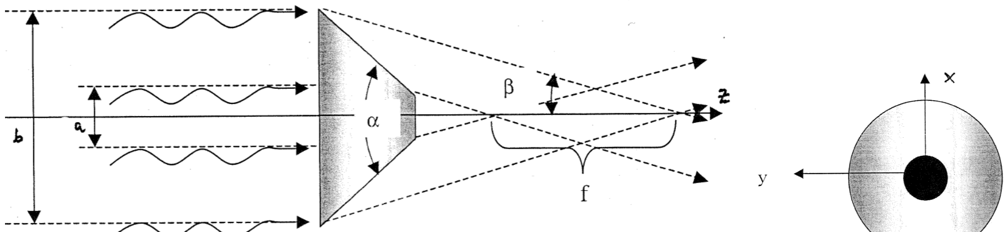


Fig. (a). Side view of the axicon. (b). View of the axicon down the z-axis.

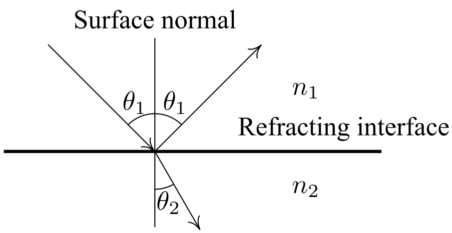
672 4. A light beam of intensity I enters a non-conducting and non-magnetic medium at normal incidence. If the index of refraction of the medium is n, what is the radiation pressure on the medium surface?

673 7. You wish to produce circularly polarized light from linearly polarized light at 589 nm . The only material available is calcite ($n_0 = 1.659, n_e = 1.487$).
(a) Calculate the thickness required to produce a true zero-order wave plate capable of producing circularly polarized light.
(b) Would quartz ($n_0 = 1.54424, n_e = 1.55336$) be better or worse for this purpose? Why?

674 7. In a prism compressor you would like to minimize the reflection of p-polarized 800 nm light in a BK7 prism ($n_{BK7} = 1.51078, n_{\text{air}} = 1.00027$). Derive a formula for Brewster's angle from the reflection factors:

$$R_p = \left(\frac{n_2/\cos\theta_2 - n_1/\cos\theta_1}{n_2/\cos\theta_2 + n_1/\cos\theta_1}\right)^2 \text{ and } R_s = \left(\frac{n_2\cos\theta_2 - n_1\cos\theta_1}{n_2\cos\theta_2 + n_1\cos\theta_1}\right)^2$$

where:



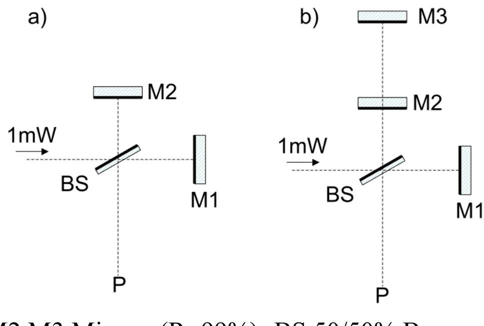
What is your incidence angle for minimum reflection? What would be the reflection loss from four reflections if you accidentally put s-polarized radiation into the system?

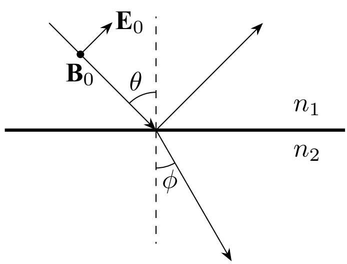
675 12. A nearly collimated femtosecond laser beam with a peak power of 10^{10} W and a beam diamater of 2 mm is incident on a gold mirror with a damage threshold of 5×10^{10} W/cm^2. You have a box containing a variety of 25 mm diameter anti-reflection coated plano-convex and plano-concave lenses with focal lengths of $\pm 50, \pm 75, \pm 100, \pm 150, \pm 200$, and ± 250 mm. Design a telescope to ensure that the beam incident on the gold mirror is collimated and below the damage threshold.

676 7. Two linear polarizers are arranged such that the angle of the transmission axis of one can be rotated with respect to the other. Randomly polarized light is incident on the two polarizers.
(a) Plot the fraction of light that is transmitted as a function of the angle between the two transmission axes.
(b) If the two polarizers have their axes orthogonal to each other, and a third polarizer is place between them midway at a 45 -degree angle, what is the transmission?
(c) Describe the effect that having $N \gg 1$ polarizers would have such that the total angular difference is 90 -degrees, but each individual polarizer is rotated by an angle of $90^\circ/$N with respect to the previous polarizer.

677 9. Professor Old designs an experiment to demonstrate the wave-like nature of light. The experiment consists of a plane-monochromatic light source (of wavelength λ) that illuminates four narrow slits at normal incidence. The slits are separated by increasing distances, $h, 2h$ and $3h$, where $h \gg \lambda$. An observation screen is placed far from the slits such that you can neglect any difference in the observation angle (θ) from the various slits.
(a) Derive an expression for the intensity profile (as a function of angle) when all four slits are equally illuminated.
(b) What is the maximum intensity and at what angles does it occur?
(c) What is the minimum intensity and at what angles does it occur?

678 7. A Michelson interferometer is constructed as in panel (a) below, with an $R = T = 50\%$ beamsplitter and two highly reflecting mirrors ($R = 99\%$). A well collimated, 1 mW HeNe laser is directed into the perfectly aligned interferometer.
(a) What is the condition that complete destructive interference is seen at point P ? Where did the incident light go?
(b) While in the condition of destructive interference, a second (parallel) mirror is added to one arm of the interferometer as shown in panel (b). The distance between the two mirrors in that arm is scanned. Plot the power seen at point P , as a function of this distance.





1-3T679

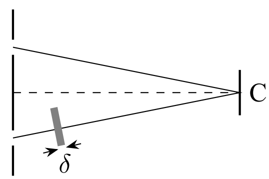
M1,M2,M3 Mirrors (R=99%). BS 50/50% Beamsplitter

679 9. Monochromatic plane waves are incident on a planar interface between two dielectric media of refractive index n_1 and n_2. Derive an expression for the reflection coefficient (ratio of the reflected field to the incident field) as a function of incidence angle, θ, for transversemagnetic (p-polarized) light. Show that there is no reflected light when $\tan(\theta) = n_2/n_1$.

680 7. (a) What type of aberration is present in lenses and absent from mirrors?
(b) For maximal polarization by reflection, what is the angle between the reflected and refracted rays?
(c) What types of phenomena provide direct experimental proof of the transverse nature of light?
(d) What is the range of frequencies for visible light?
(e) For a given thick lens, what condition must a ray not parallel to the axis satisfy to go through the lens undeviated?
(f) A double slit is illuminated first by red light and then by violet light. Which color gives the wider interference pattern beyond the double slit? Justify your answer.

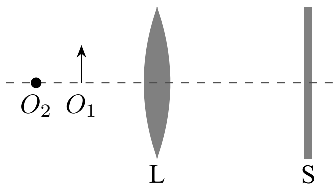
681 11. A "single-mode" optical fiber actually supports two field modes that have nearidentical, near-Gaussian intensity distributions but orthogonal linear polarizations. The refractive indices for the two modes may differ slightly due to some deviation of the fiber profile from cylindrical symmetry. Assume that a piece of such a fiber extends along the z axis from $z = 0$ to $z = L$, where L is the fiber length, and that the field modes correspond to polarizations along the x and y-directions, with respective refractive indices of $n_x = 1.4475$ and $n_y = 1.4476$. The fiber piece is used to turn linearly polarized light (wavelength 852 nm) into circularly polarized light. The light enters the fiber at $z = 0$ and exits at $z = L$.
(a) Specify all polarization directions (between 0 and π) of the incident light with respect to the x-axis and all possible fiber lengths that produce circularly polarized light.
(b) Identify the solution that will produce left-circularly polarized light for the shortest possible fiber length.

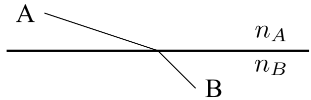
682 7. The figure below shows a double slit experiment in which monochromatic light of wavelength λ from a distant source is incident upon two slits, each of width $w (w \ll \lambda)$, and the interference pattern is viewed on a distant screen. A thin piece of glass of thickness δ and index of refraction n, is placed between one of the slits and the screen. and the intensity at the central point C is measured as a function of δ. The intensity for $\delta = 0$ is given by I_0. (Assume that the glass does not absorb any light)
(a) What is the intensity at point C as a function of thickness δ ?
(b) For what values of δ is the intensity at C a minimum?
(c) Suppose that the width of one of the slits is now increased to $2w$, the other width remaining unchanged. What is now the intensity at point C as a function of δ ?



683 11. The University of Michigan is planning to launch a satellite for a physics experiment. The satellite will have a geostationary orbit and will send signals to Ann Arbor at a frequency of 15 GHz . Estimate the size of the parabolic antenna needed to direct the signals approximately to a circular region of radius 300 km around the Physics Department. [Note: The mass of earth is $M_E = 5.97 \times 10^{24}$ kg.]

684 8. A converging lens (labeled L) focuses the image of object 1 (O_1) on the screen (labeled S).
(a) Use ray optics to locate the image of O_2 in the figure above. You are free to choose an appropriate focal length.
(b) Determine the size of the spot created by O_2 on S in terms of ℓ (where ℓ is the distance of O_1 from L), f the focal length of the lens, and D the diameter of the lens.
(c) Suppose that S is a CCD sensor array like those found in a modern camera. Estimate the maximum separation between O_1 and O_2 for which both objects are in focus. Explain your reasoning.





686 8. (a) Consider points A and B which lie on two sides of a plane boundary separating regions of uniform index of refraction n_A and n_B, as shown below. A light ray travels between A and B . The ray will consist of straight line segments in the two media, but the segments in regions n_A and n_B will travel in different directions owing to refraction at the interface. Show that the path given by the law of reflection (Snell's law) is the one that takes the least time for passage from point A to point B. The statement that light rays follow the path of least time of propagation is known as *Fermat's Principle*.
(b) Interference fringes are produced by thin wedge-shaped film of plastic of refractive index 1.4. If the angle of the wedge is 20 arc seconds and the distance between fringes is 0.25 cm , find the wavelength of the (monochromatic) light incident perpendicular on the plastic. (NOTE on anglular units: One arc second is $1/60$ th of an arc minute, and one arc minute is $1/60$ th of a degree.)

687 12. Consider a scalar optical field $U(\mathbf{r})$ with source $S(\mathbf{r})$ which propagates in free space and obeys the reduced wave equation

$$\left(\nabla^2 + k_0^2\right) U(\mathbf{r}) = -4\pi S(\mathbf{r}),$$

where k_0 is the wave number. (a) Show that in the far zone that $U(\mathbf{r})$ behaves asymptotically as an outgoing spherical wave with

$$U(\mathbf{r}) \sim \frac{e^{ik_0 r}}{r} \hat{S}(\mathbf{k})$$

where $\mathbf{k} = k_0 \hat{\mathbf{r}}$ and $\hat{S}(\mathbf{k})$ denotes the Fourier transform of $S(\mathbf{r})$, which is defined by

$$\hat{S}(\mathbf{k}) = \int d^3 r e^{-i\mathbf{k}\cdot\mathbf{r}} S(\mathbf{r})$$

(b) Use this result to show that

$$S_{\mathrm{NR}}(\mathbf{r}) = \left(\nabla^2 + k_0^2\right) F(\mathbf{r}),$$

where $F(\mathbf{r})$ is an arbitrary function, is a source which produces a field $U(\mathbf{r})$ which vanishes asymptotically in the far zone. Such a source is referred to as nonradiating.

688 8. A soap bubble produces constructive interference in the reflected light when illuminated by light whose wavelength in air is 600 nm . The index of refraction of the soap bubble is 1.33 . What is the minimum thickness of the bubble?

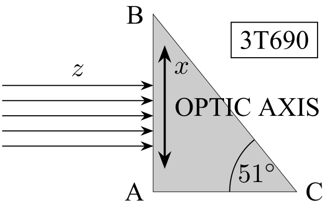
689 10. A person has a far point of infinity and a near point of 83 cm . What is the refractive power of contact lenses this person should wear in order to be able to read a cell phone screen located 25 cm from her eyes?

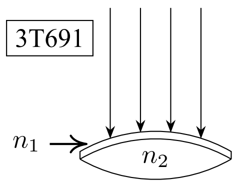
690 8. A beam of monochromatic ($\lambda = 0.5893\,\mu$m) and linearly polarized light moving along the $\hat{z}$ axis, reaches normally the surface AB of a prism of calcite ($n_0 = 1.6584$ and $n_e = 1.4864$). The electric field subtends an angle θ with the plane of incidence. The optical axis is along the $\hat{x}$ axis, parallel to the surface AB (see fig). Find what happens to the beam through the prism and when it emerges from the surface BC.

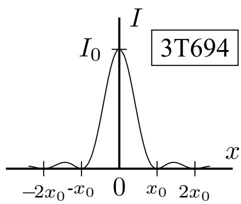
691 12. A lens of refractive index $n_2 = 1.85$ is covered on the top surface with a very thin layer of a transparent substance, whose refractive index is n_1 and thickness is t. A beam of monochromatic light ($\lambda = 0.4\mu$ m) is incident normally on the layer and the lens (see figure). Assuming that the reflectivities are the same for the air-layer and layer-lens boundary surfaces, find the values n_1 and t for which the reflected ray is subject to destructive interference.

692 8. Light of wavelength λ and intensity I is incident perpendicularly onto an opaque circular disk of radius R. Find the intensity of light at a point P on the line perpendicular to the disk and passing through its center. You may assume that the distance between P and the center of the disk is much larger than R.

693 12. A glass cube has a refractive index of 1.5. A beam of light enters the top face obliquely and then strikes the side of the cube. Does any light emerge from the side?



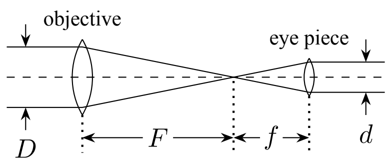




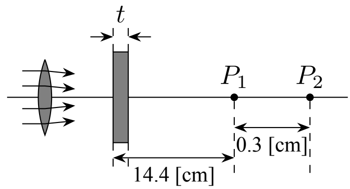
694 **I-2. Diffraction** A long thin silt is illuminated from behind by a monochromatic source of light. The figure shows the resulting intensity pattern on a screen in the far field region. The pattern changes when the right half of the width of the slit is covered by an absorptive filter which decreases the transmitted electric field in that region by a factor of two. Find the new intensity at the three locations $x = 0, x = x_0$, and $x = 2x_0$ in terms of I_0, the maximum intensity in the original pattern. [Hint: use a graphical method of adding complex E fields. Fourier techniques are not necessary.]

695 Propose a model to explain why the sky is blue and justify your reasoning with a simple analytic calculation. [Note: the atmosphere is not made up of free electrons!]

696 **2. Telescope** a) Find the magnification of the Keplerian telescope shown above. Express your answer in terms of the quantities shown in the figure.
b) Estimate the resolution of such a telescope.
c) Often in telescopes such as this, each of the two lenses is made up of two different optical materials. Explain why this is done.



697 8. Rays from a lens are converging toward a point image P_1 located to the right of the lens. What thickness t of glass with index of refraction 1.60 must be interposed between the lens and P_1 for the image to be formed at P_2, located 0.3 cm to the right of P_1 ? The locations of the piece of glass and of points P_1 and P_2 are shown in the figure.



698 12. A continuous, collimated Gaussian beam with wavelength $\lambda = 532$ nm, w-parameter (waist size) of 5 mm , and total power 1 W is to be focused into a small experimental region such that the peak intensity at the center of the focal spot is $I_0 = 10^7 \frac{\mathrm{W}}{\mathrm{cm}^2}$. You are supposed to pick a lens for the job. Assume that all light power passes through the aperture of the lens.
(a) What is the central (peak) intensity of the beam before it enters the lens?
(b) What is the beam waist size w at the focal spot?
(b) What focal length should you pick for the lens? (Assume an ideal lens without aberrations).
Useful equations: (r = distance from beam axis, z = distance from focus)
For given w the intensity follows $I(r) = I_0 \exp\left(\frac{-2r^2}{w^2}\right)$.

Near a focal spot, $w(z) = w_0\sqrt{1 + \left(\frac{z}{z_R}\right)^2}$, where the Rayleigh length $z_R = \frac{\pi w_0^2}{\lambda}$ and w_0 is the w-parameter (waist size) at the focus. Gaussian integrals (see front of the exam).

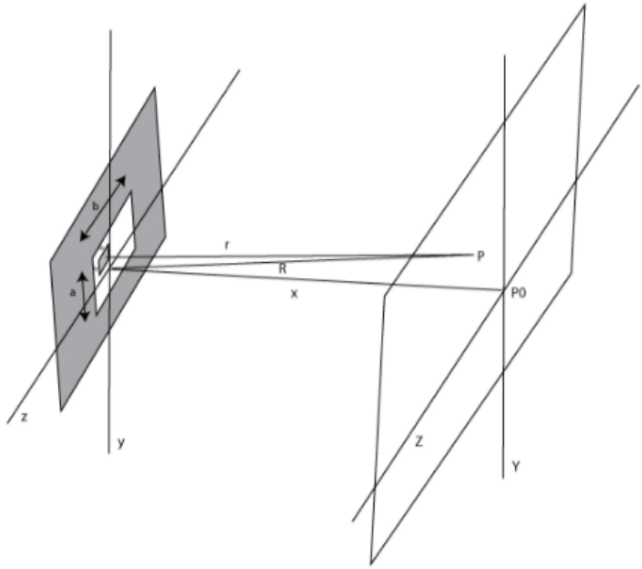
699 8. A lens of positive focal length f is placed at the origin $x = 0$, and another lens of negative focal length $-f$ is placed at position $x = d$ along the x-axis. If an object P is placed on the x-axis a distance s from the first lens ($x = -s$), and assuming f, f', d, and s are all positive,
a) compute the image position of light from the object going through the two-lens system.
From this general expression find the image position in the special circumstances where
b) $s = \infty$ and $f = d$,
c) $s = f = f' = d$, and
d) $s = 4d$, $f = d$ and $f' = 2d$.

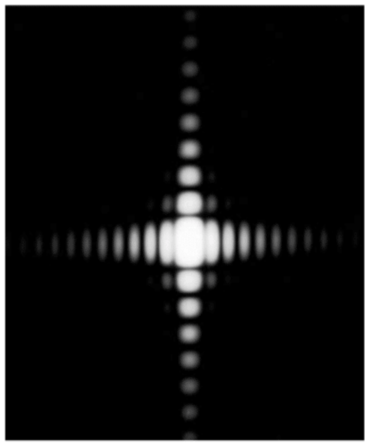
700 10. A piece of clear scotch tape is stretched with about half the force that would be required to rip it. After the stretching the piece of tape is found to act as a (zero-order) quarter-wave plate when light is passed through it at normal incidence. It is further observed that the fast axis (lower n) and the slow axis (larger n) are parallel and transverse to the stretching (but we don't know yet which axis corresponds to which direction).
(a) Assuming a test wavelength of $\lambda = 532$ nm and a tape thickness of $d = 0.25$ mm, determine the difference in refractive index between the fast and slow axes that results from the stretching.
(b) Noting that the tape contains a polymer plastic material, determine which direction on the tape (parallel or transverse to the stretching) corresponds to what axis (fast or slow). You must explain your reasoning.
(c) Assume that the stretching direction is along the x-axis, that the light beam propagates in the positive z-direction, and that the incident polarization vector is $\frac{1}{\sqrt{2}}(\hat{\mathbf{x}} + \hat{\mathbf{y}})$. Is the circular polarization of the transmitted light beam left- or right-handed?

Instruction for part c): If you cannot do part b), assume in part c) that the stretching direction corresponds to the slow axis (larger n). In case you need to follow this instruction you must clearly state this. This instruction may or may not reflect the truth, so don't take it as a hint for part b).

Reminder: If you look into an oncoming beam of circularly polarized light and the tip of the electric-field vector rotates counterclockwise, the beam is right-handed.

701 2. A laser beam is incident from the left (pointed in the $+\hat{x}$ direction) upon a rectangular aperture with length a in the y-direction and length b in the z-direction, where $b > a$ (see Figure). The laser is monochromatic with a wavelength λ and its intensity is uniform across the opening. Light passing through the aperture is collected on a screen at a very large distance $x(\gg a, b, \lambda)$ away from both the aperture and the laser. Coordinates on the distant screen are denoted by their physical coordinates (Y, Z), or by the angles ($\theta_Y \approx Y/x, \theta_Z \approx Z/x$) measured with respect to the $\hat{x}$ axis.





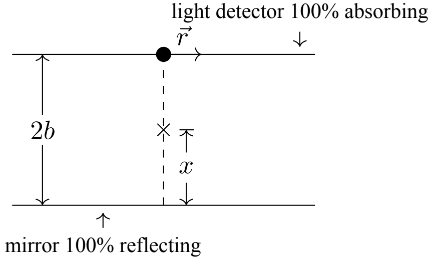
(a) The intensity pattern $I\left(\theta_Y, \theta_Z\right)$ as measured on the distant screen is shown in the Figure on the right. Which directions in this figure correspond to the Y axis, and which correspond to the Z axis? (example answer: the vertical direction corresponds to the Y axis) Explain your reasoning.
(b) How does the angular distribution of intensity $I\left(\theta_Y, \theta_Z\right)$ change if the screen is moved a factor of 2 further away from the aperture? Explain your answer.
(c) How would the measured intensity distribution $I\left(\theta_Y, \theta_Z\right)$ change if the wavelength of the laser light λ is doubled? Briefly explain your answer.
(d) Calculate the intensity distribution $I\left(\theta_Y, \theta_Z\right)$ in terms of the angles θ_Y, θ_Z and normalized to the intensity $I(0)$ at the center of the screen. Hint: Do not worry about the constant amplitude in front of the electric field or intensity until the end of the derivation.

702 3. A point dipole source of monochromatic light (wavelength λ) is suspended a height x above a perfectly reflecting mirror. Emitted light directed down towards the mirror below has the same amplitude and polarization as light emitted upwards in the opposite direction. A planar, fullyabsorbing light detector, oriented exactly parallel to the mirror is placed a height $2b$ above the mirror. The detected light intensity depends on the source position x and on r, the distance from an axis through the source and normal to the mirror and detector, as shown in the figure. Assume the light emitter is small enough that it intercepts none of the reflected light. Assume in what follows that $\lambda \ll b$ and $r \ll 2b - x$. There are many local maxima at $r = 0$ as x is varied, and at any fixed x as r is varied.

(a) At what values of x are intensity maxima observed on the light detector at $r = 0$?

(b) The source is placed at $x = b$ and the wavelength λ is such that an intensity maximum is observed on the light detector at $r = 0$. At what other values of r will intensity maxima be observed on the light detector?

(c) How is the answer to part (a) changed if the mirror moves with velocity v parallel to the detector plane? How is the answer to part (b) changed?



703 2. For a system consisting of two thin lenses, the image distance after the second lens is

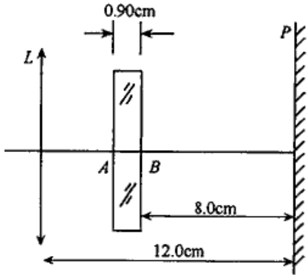
$$s_{i_2} = \frac{f_2 d - (f_1 f_2 s_{o_1}) / (s_{o_1} - f_1)}{d - f_2 - (f_1 s_{o_1}) / (s_{o_1} - f_1)}$$

Here, f_1 and f_2 are the focal lengths of the two lenses, d is the distance between the two lenses, and s_{o_1} is the object distance in front of first lens. Derive expressions for (i) the back focal length, (ii) the front focal length, and (iii) the effective focal length of the two-lens system as $d \to 0$.

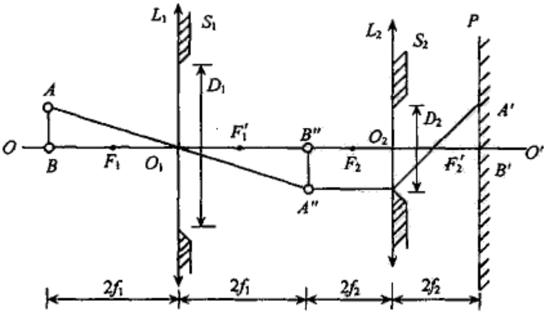
740 1.63 照相机镜头 L 前 2.28 m 处的物体被清晰地成像在镜头后面 12.0 cm 处的照相胶片 P 上．今将一折射率为 1.50 、厚 $AB = 0.90$ cm 、两面平行的玻璃平板插入镜头与胶片之间，与光轴垂直，如图 1 所示．设照相机镜头可看作一个简单薄凸透镜，光线为近轴光线.

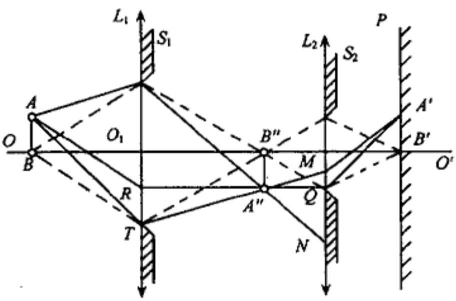
（1）求插入玻璃板后，像的新位置.

（2）如果保持镜头、玻璃板、胶片三者间距离不变，并要求物体仍然清晰地成像于胶片上，则物体应放在何处?



742 1.67 如图 1 所示，L_1 和 L_2 为两个共轴薄透镜，OO' 为主光轴，L_1 和 L_2 的焦距分别为 f_1 和 f_2 ，紧靠薄透镜 L_1 和 L_2 有光圈 S_1 和 S_2 ，光圈透光部分直径分别为 D_1 和 D_2 . 两薄透镜相距 $2(f_1 + f_2)$ ，透镜 L_2 右方相距 $2f_2$ 处有一光屏 P（垂直于 OO' ）。A 、B 为亮度相等的两光点，B 点在光轴 OO' 上，离 L_1 距离为 $2f_1, AB$ 垂直 $OO', AB = a$ ．又已知 $D_1 = 4a$ ，$D_2 = 2a$ ，并设 A 、B 点对 L_1 光圈透光孔所张的角相等．设 A 、B 在 P 上成像分别为 A' 和 B' .





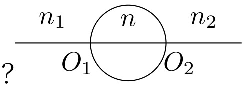
（1）求 A' 和 B' 亮度的比值（此小题设 $f_1 = 2f_2$ ）;

（2）若 A 、B 点，以及 L_1 、L_2 、屏 P 位置不变，

(i) 为使 A' 像消失，光圈 S_1 的孔径 D_1 最大为多大（光圈 D_2 不变时）;
(ii) 光圈 D_1 和 D_2 均保持原值不变，并让像 A' 和 B' 亮度相等，但不改变像的位置，可在 OO' 上插放一同轴薄凸透镜 L_3，求 L_3 应放在何处？镜片直径至少要多大？焦距应为多少？

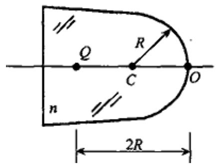
744 1.72 薄壁玻璃球内充满 $n = 4/3$ 的水．观察者沿直径方向看沿直径从远端逐渐移近的物体．设球半径 $r = 5\ \mathrm{cm}$，移动速率为 $v = 1\ \mathrm{cm/s}$．试分析像的位置如何变化，并计算物体开始时以及移动了 2.5 cm、5.0 cm 时像的速率．

746 1.76 如图所示，一个玻璃球，$n = 1.5$，半径为 R，置于空气中，问



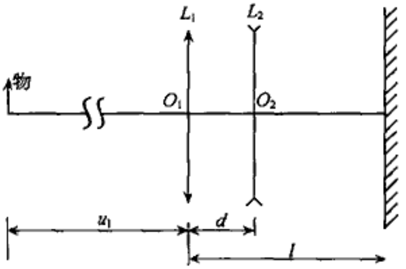
（1）物在无限远时经球成像于何处？
（2）物距球前表面 $2R$ 处时成像于何处，大小如何？
(3)如果物是一个指向球面的小箭头,在(2)条件下成像的箭头指向何方？

748 1.79 体温计横截面如图所示．已知细水银柱 Q 离圆柱面顶点 O 的距离为 $2R, R$ 为该柱面的半径，C 为圆柱面中心轴位置，玻璃折射率 $n = \dfrac{3}{2}$．如果人在右方观察，试求：在图示横截面上观察者从右方所看到的水银柱像的位置及其大小．



750 1.82 一个光学系统结构如图所示，一个薄凸透镜 L_1 焦距为 f_1，另一个薄凹透镜 L_2 焦距为 $-f_2$．成像面 P 处放有照相底片，L_1 与 P 的位置固定不动．现给定 $f_1 = 3.00\ \mathrm{cm}, P$ 与 L_1 之间距离 $l = 4.50\ \mathrm{cm}. L_1$ 和 L_2 之间距离 d 是可调的．要求通过调节 d 使无穷远处的物或近处的物都能在底片上成实像，问：

（1）如果 $f_2 = 3.00\ \mathrm{cm}$，物体从无穷远处移到 $u_1 = 100.0\ \mathrm{cm}$ 处，则 L_2 移动的距离应为多少？
（2）是否只要 f_2 和 d 取值适当，不管物体在什么地方都能在 P 上成实像？如不能，则对物距有何限制？
（3）如果要求采用一个焦距确定的 L_2，通过调节 d 的数值使物距满足上面（2）中要求的物体都能在 P 上成实像，则 L_2 的焦距 f_2 应满足什么条件及相应的 d 的调节范围．

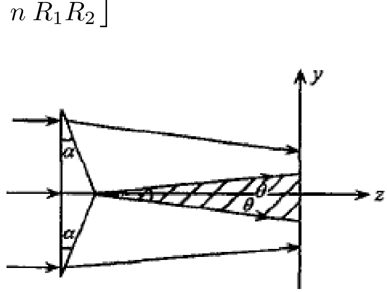


752 1.86 试推导对于任何光谱段，空气中厚透镜消焦距色差的条件．已知厚度为 d，曲率半径分别为 R_1 和 R_2 的厚透镜的光焦度公式为

$$\Phi = (n-1)\left[\left(\frac{1}{R_1} + \frac{1}{R_2}\right) - \frac{d}{n}\frac{n-1}{R_1 R_2}\right]$$

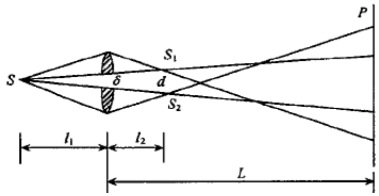
754 2.4 证明平面电磁波公式 $E = A\cos(\omega t - kx)$ 是波动微分方程 $\dfrac{\partial^2 E}{\partial x^2} - \dfrac{1}{v^2}\dfrac{\partial^2 E}{\partial t^2} = 0$ 的解．

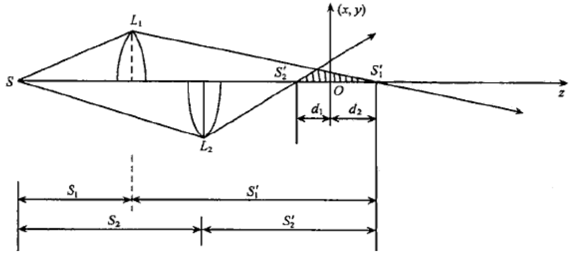
756 2.7 波长 $\lambda = 6000\mathrm{\mathring{A}}$ 的单色平行光正入射到菲涅耳双棱镜的底面，棱镜底角 $\alpha \ll 1^\circ$，折射率 $n = 1.5$．
（1）出射光会在屏幕上形成什么图样，屏幕平行棱镜底面．
（2）若屏上条纹间距为 0.1 mm，求棱镜底角 α．



758 2.10 焦距为 f 的透镜前相距 $l_1\,(l_1 > f)$ 处放置一单色缝光源（波长为 λ），透镜后相距 L 处放置一观察屏 P，整个系统沿主光轴对称．现沿平行于线光源方向把透镜对

称地切割成两部分，并沿切口垂直方向对称地移开一小段距离 δ ，这个装置称比累透镜，如图所示．求屏 P 上干涉条纹间距和条纹总数．



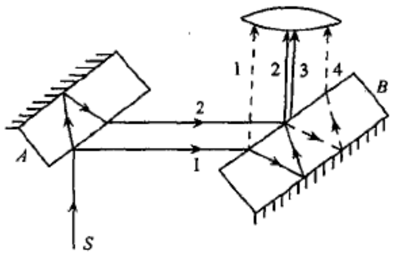


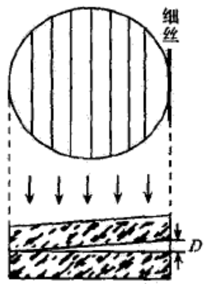
760 2.13 如图所示为梅斯林干涉装置．将一凸透镜沿直径对切后沿光轴错开一定距离，单色点光源 S 位于光轴上，经上下两个半透镜成像于 S'_1 和 S'_2 ，在两束相干光重叠区内，垂直于光轴设观察屏．

（1）求屏上干涉条纹的形状．

（2）设透镜焦距 $f=30$ cm ，点源距较近的半透镜 L_1 为 60 cm ，两个半透镜错开 8 cm ，单色光波长 500 nm ，屏幕放在 S'_1 和 S'_2 的中点，求旁轴条件下干涉条纹的间距．

762 2.15 在焦距 $f=10$ cm 的物镜焦平面上观察雅满干涉仪产生的花样，如图．干涉仪两板用折射率 $n=1.5$ 的玻璃制成，厚度 $h=2$ cm ，两板间夹角 $\alpha=1'$ ．光射到板上，入射角 45° ．观察到后焦面上条纹间距 $\Delta x=3.84$ mm ．求光波波长．





764 2.17 在单色光杨氏双缝实验中的一个光路中放置玻璃片，其折射率为 n ，厚度为 t ．求屏中心处光强；t 取什么值时，中心处光强最小；又若一缝宽是另一缝的 2 倍，上述情况又如何？设 $t=0$ 时中心处光强为 I ．忽略玻璃的吸收．

766 2.20 块规是机加工里用的一种长度标准，它是一钢质长方体，它的两个端面经过磨平抛光，达到相互平行．附图中 G_1 ，G_2 是同规号的两个块规，G_1 的长度是标准的，G_2 是要校准的. 校准方法如下: 把 G_1 和 G_2 放在钢质平台面上, 使面和面严密接触, G_1, G_2 上面用一块透明平板 T 压住．如果 G_1 和 G_2 的高度（即长度）不等，微有差别，则在 T 和 G_1, G_2 之间分别形成尖劈形空气层，它们在单色光照射下各产生等厚干涉条纹．

（1）设入射光的波长是 5893Å, G_1 和 G_2 相隔 5 cm （即图中的 l），T 和 G_1, G_2 间干涉条纹的间距都是 0.5 mm ，试求 G_2 和 G_1 的高度之差．怎样判断它们谁长谁短？

（2）如果 T 和 G_1 间干涉条纹的间距是 0.5 mm ，而 T 和 G_2 间的是 0.3 mm ，则说明什么问题？

768 2.23 将曲率半径 1 m 的薄凸透镜贴在平晶上，钠光（5893Å ）垂直照明，在反射光中观察牛顿环．然后在球面和平晶间充满四氯化碳 $(n=1.461)$ ．求充液前后第 5 个暗环的半径比，充液后的第 5 个暗环半径是多少？